Chapter 3 — Cells and Subcellular Organelles

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Fluid mosaic model was given by: (a) Singer-Nicolson (b) Hooke-Brown (c) Watson-Crick (d) Schleiden-Schwann
2. Cholesterol in membranes helps to: (a) Store DNA (b) Make proteins (c) Form cell walls (d) Maintain fluidity
3. Ribosomal subunits are assembled in: (a) Golgi apparatus (b) Nucleolus (c) Lysosome (d) Peroxisome
4. Rough ER is mainly involved in: (a) ATP synthesis (b) Lipid synthesis (c) Protein synthesis (d) Detoxification
5. Inner mitochondrial membrane infoldings are: (a) Grana (b) Cisternae (c) Thylakoids (d) Cristae
6. Golgi face near the ER is: (a) Cis face (b) Trans face (c) Catalytic face (d) Binding face
7. Active transport is: (a) Channels only (b) Along gradient, no ATP (c) Against gradient, using ATP (d) Vesicles only
8. Totipotent stem cells can form: (a) Only nerve cells (b) Only muscle cells (c) All cell types (d) Only blood cells

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Why is the plasma membrane described as fluid and mosaic?
2. Why are mitochondria and chloroplasts considered semi-autonomous organelles?
3. How does the Golgi apparatus modify and transport cellular products?
4. Why are lysosomes called the digestive compartments of the cell?
5. Differentiate between osmosis and diffusion.
6. How are induced pluripotent stem cells useful in regenerative medicine?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Explain the fluid mosaic model of the plasma membrane. **(4)**
(b) Describe the structure and functions of mitochondria. **(4)**
Q.4 (a) Describe the mechanisms of membrane transport. **(4)**
(b) Describe the categories of stem cells and their uses in medicine. **(4)**

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